Radiation of charged particle

$$\frac{dP}{d\Omega} = \frac{\mu_0}{16\pi^2 c}\left(\ddot{\vec{d}} \times \hat{n}\right)^2 = \frac{\mu_0}{16\pi^2 c}\,\ddot{\vec{d}}^{\,2}\sin^2\theta$$

Total power $P = \int \frac{\mu_0}{16\pi^2 c}\,\ddot{\vec{d}}^{\,2}\sin^2\theta\, d\varphi \sin\theta\, d\theta$

$$\int_0^{\pi} \sin^2\theta \sin\theta\, d\theta = -\int_1^{-1}(1-\cos^2\theta)\, d(\cos\theta) = \left(x - \tfrac{1}{3}x^3\right)\Big|_{-1}^{1} = 2\left(1-\tfrac{1}{3}\right) = \tfrac{4}{3}$$

$$P = \frac{\mu_0 q^2}{16\pi^2 c}\,\ddot{\vec{r}}^{\,2}\,\frac{2\pi \cdot 4}{3} = \frac{\mu_0 q^2}{6\pi c}\,\dot{\vec{v}}^{\,2} = \frac{\mu_0 q^2}{6\pi c}\,\vec{a}^{\,2}$$

this the rate of energy loss by the particle!

Expression for dipole radiation, on which this result is based, is approximate: we neglected $\frac{\hat{r}\cdot\vec{r}'}{c}$ term in $t_r = t - \frac{r}{c} + \frac{\hat{r}\cdot\vec{r}'}{c}$ in the argument of $\vec{j}(\vec{r}', t_r)$.

Suppose charges move with speed v in volume of size l. Then charge distribution changes on the scale $T \sim \frac{l}{v}$.

We can neglect $\frac{\hat{r}\cdot\vec{r}'}{c} \sim \frac{l}{c}$ if it is much smaller than T:

$$\frac{l}{c} \ll \frac{l}{v} \Rightarrow \boxed{v \ll c}$$ non-relativistic limit!

The same condition $\frac{l}{c} \ll T \Rightarrow l \ll cT$

But $T \sim \frac{1}{\omega}$ where ω is the frequency of emitted radiation

$$\Rightarrow l \ll \frac{c}{\omega} = \frac{c}{ck} = \lambda$$

$$\boxed{l \ll \lambda}$$ size much smaller than the wavelength.

What if $v \sim c$?

Relativistic generalization of energy loss f-la

In rest-frame of moving charge $(\vec{v}=0, \vec{a}\neq 0)$

$$\frac{d\mathcal{E}}{dt} = dI = \frac{\mu_0}{6\pi c} q^2 a^2$$

Note that emitted (radiated) momentum $\underline{d\vec{P}=0}$!

Consider $dp^i \sim \frac{du^k}{ds}\frac{du_k}{ds}\, dx^i = w^k w_k\, dx^i$

4-velocity $\quad u^k = \frac{dx^k}{ds} = \left(\frac{c\,dt}{c\,dt/\gamma}, \frac{d\vec{x}}{dt\, c/\gamma}\right) = \gamma\left(1, \frac{\vec{v}}{c}\right)$

$$ds = c\,dt\sqrt{1-v^2/c^2} = c\,dt/\gamma = c\,d\tau$$

4-acceleration $\quad w^k = \frac{du^k}{ds} = \frac{\gamma}{c}\frac{d}{dt}\left(\gamma, \frac{\gamma}{c}\vec{v}\right)$

$$\frac{d}{dt}\gamma = \frac{d}{dt}\frac{1}{\sqrt{1-v^2/c^2}} = -\frac{-2\,\vec{v}\cdot\dot{\vec{v}}/c^2}{2\left(1-v^2/c^2\right)^{3/2}} = \gamma^3\frac{\vec{v}\cdot\dot{\vec{v}}}{c^2}$$

$$\frac{d}{dt}\left(\gamma\vec{v}\right) = \gamma^3\frac{(\vec{v}\cdot\dot{\vec{v}})}{c^2}\vec{v} + \gamma\dot{\vec{v}}$$

$$\Rightarrow w^k = \left(\gamma^4\frac{\vec{v}\cdot\dot{\vec{v}}}{c^3}, \gamma^2\frac{\dot{\vec{v}}}{c^2} + \gamma^4\frac{\vec{v}(\vec{v}\cdot\dot{\vec{v}})}{c^4}\right)$$

check that $\quad w^k u_k = \gamma^5\frac{\vec{v}\cdot\dot{\vec{v}}}{c^3} - \frac{\gamma\vec{v}}{c}\left(\frac{\gamma^2}{c^2}\dot{\vec{v}} + \frac{\gamma^4}{c^4}\vec{v}(\vec{v}\cdot\dot{\vec{v}})\right)$

$$= \frac{\gamma^5}{c^3}\vec{v}\cdot\dot{\vec{v}} - \frac{\gamma^3}{c^3}\vec{v}\cdot\dot{\vec{v}} - \frac{\gamma^5}{c^5}v^2(\vec{v}\cdot\dot{\vec{v}}) =$$

$$\frac{\gamma^3}{c^3}\underbrace{\left(1+\frac{v^2}{c^2}\gamma^2\right)}\vec{v}\cdot\dot{\vec{v}}$$

$$1+\frac{v^2/c^2}{1-v^2/c^2} = \frac{1}{1-v^2/c^2} = \gamma^2$$

$$= \frac{\gamma^5}{c^3}\vec{v}\cdot\dot{\vec{v}} - \frac{\gamma^3}{c^3}\left(\gamma^2\vec{v}\cdot\dot{\vec{v}}\right) = 0$$

Now, in the rest frame $\vec{v}=0$, $\gamma=1$

$$w^k = \left(0, \frac{\dot{\vec{v}}}{c^2}\right) = \left(0, \frac{\vec{a}}{c^2}\right)$$

$$\Rightarrow \quad w^k w_k = -\frac{a^2}{c^4}$$

$$w^k w_k \, dx^0 = \left(-\frac{a^2}{c^4}\right) c\,dt = -\frac{a^2}{c^3}\,dt$$

Hence we can write $\Big\}$ $$d\mathcal{E} = \frac{\mu_0 q^2}{6\pi c} a^2 dt = \frac{\mu_0 q^2}{6\pi c}\left(-c^3 w^k w_k \, dx^0\right)$$

$$dp^i = \frac{1}{c}\cdot\frac{\mu_0 q^2}{6\pi c}\left(-c^3 w^k w_k \, dx^i\right) \qquad p^i = \left(\frac{\mathcal{E}}{c}, \vec{p}\right)$$

Now, In arbitrary frame

$$c^3 w^k w_k = \frac{\gamma^8}{c^3}(\vec{v}\cdot\vec{a})^2 - c^3\left(\frac{\gamma^2}{c^2}\vec{a} + \frac{\gamma^4}{c^4}\vec{v}(\vec{v}\cdot\vec{a})\right)^2 =$$

$$= \frac{\gamma^8}{c^3}(\vec{v}\cdot\vec{a})^2 - c^3\left(\frac{\gamma^4}{c^4}a^2 + \frac{\gamma^8}{c^8}v^2(\vec{v}\cdot\vec{a})^2 + \frac{2\gamma^6}{c^6}(\vec{v}\cdot\vec{a})^2\right)$$

$$= \frac{\gamma^8}{c^3}(\vec{v}\cdot\vec{a})^2 - \left(\frac{\gamma^4}{c}a^2 + \frac{c^3\gamma^6}{c^6}(\vec{v}\cdot\vec{a})^2\left[2 + \frac{\gamma^2 v^2}{c^2}\right]\right) =$$

$$2 + \frac{v^2/c^2}{1-v^2/c^2} = \frac{2-v^2/c^2}{1-v^2/c^2}$$

$$= \frac{\gamma^8}{c^3}(\vec{v}\cdot\vec{a})^2 - \left(\frac{\gamma^4}{c}a^2 + \frac{\gamma^6}{c^3}(\vec{v}\cdot\vec{a})^2\left[\gamma^2+1\right]\right) \qquad \gamma^2\left(1+\frac{1}{\gamma^2}\right) = \gamma^2+1$$

$$= -\left(\frac{\gamma^4}{c}a^2 + \frac{\gamma^6}{c^3}(\vec{v}\cdot\vec{a})^2\right)$$

$$(\vec{v}\times\vec{a})^2 = v^2a^2\sin^2\theta = v^2a^2(1-\cos^2\theta) = \vec{v}^2\vec{a}^2 - (\vec{v}\cdot\vec{a})^2$$

$$\cancel{\frac{\gamma^4}{c}\left(a^2 + \frac{\gamma^6}{c^3}\left[\vec{v}^2 - (\vec{v}\times\vec{a})^2\right]\right.}$$

$$\frac{\gamma^6}{c^3}\left[\frac{c^2}{\gamma^2}a^2 + (\vec{v}\cdot\vec{a})^2\right] = \frac{\gamma^6}{c^3}\left[c^2a^2\left(1-\frac{v^2}{c^2}\right) + (\vec{v}\cdot\vec{a})^2\right]$$

$$= \frac{\gamma^6}{c^3}\left\{c^2a^2 - \left[v^2a^2 - (\vec{v}\cdot\vec{a})^2\right]\right\} =$$

$$= \frac{\gamma^6}{c}\left\{a^2 - \frac{(\vec{v}\times\vec{a})^2}{c^2}\right\}$$

because $p^i = \left(\frac{\varepsilon}{c}, \vec{p}\right)$

$$\Rightarrow \quad dp^i = \left(\frac{1}{c}\right)\gamma^6\left[a^2 - \left(\vec{a}\times\frac{\vec{v}}{c}\right)^2\right]\frac{dx^i}{c}\left(\frac{\mu_0 q^2}{6\pi c}\right)$$

$$\Rightarrow \quad d\left(\frac{\varepsilon}{c}\right) = \left(\frac{1}{c}\right)\frac{\mu_0 q^2}{6\pi c}\,\frac{a^2 - (\vec{a}\times\vec{v})^2/c^2}{(1-v^2/c^2)^3}\,\frac{c\,dt}{c}$$

(radiated energy: relativistic result)

$$d\vec{p} = \frac{\mu_0 q^2}{6\pi c^2}\,\frac{a^2 - (\vec{a}\times\vec{v})^2/c^2}{(1-v^2/c^2)^3}\,\frac{\vec{v}}{c}\,dt \qquad \left(\frac{d\vec{x}}{c} = \frac{\vec{v}}{c}\,dt\right)$$

$$\mu_0 = \frac{1}{c^2\varepsilon_0}$$

$$\Rightarrow \quad d\vec{p} = \frac{\vec{v}}{c^2}\,d\varepsilon$$

$$d\vec{p} = \frac{\mu_0 q^2}{6\pi c^2}\,\frac{\vec{a}^2 - (\vec{a}\times\vec{v})^2/c^2}{(1-v^2/c^2)^3}\,\frac{\vec{v}\,dt}{c^2} = \frac{\vec{v}}{c}\,d\left(\frac{\varepsilon}{c}\right) = \frac{\vec{v}}{c^2}\,d\varepsilon$$

$$\frac{d\varepsilon}{dt} = \frac{\mu_0 q^2}{6\pi c}\,\gamma^6\left[a^2 - \left(\frac{\vec{a}\times\vec{v}}{c}\right)^2\right].$$

Angular distribution

$$\vec{E} = \frac{q}{4\pi\varepsilon_0}\left\{ \frac{1-v^2/c^2}{\left(R - \frac{\vec{R}\cdot\vec{v}}{c}\right)^3}\left(\vec{R} - \frac{\vec{v}}{c}R\right) + \frac{\vec{R}\times\left[\left(\vec{R}-\frac{\vec{v}}{c}R\right)\times\vec{a}\right]}{c^2\left(R - \frac{\vec{v}\cdot\vec{R}}{c}\right)^3}\right\}$$

$$\vec{R} = \vec{r} - \vec{r}(t') \qquad t' = \text{retarded time}$$

Large distances $\vec{R} = \hat{n}R$ need only radiation $\left(\frac{1}{R}\right)$ part

$$\vec{E} \approx \frac{q}{4\pi\varepsilon_0}\frac{1}{c^2R}\frac{\hat{n}\times\left[\left(\hat{n}-\frac{\vec{v}}{c}\right)\times\vec{a}\right]}{\left(1-\frac{\vec{v}\cdot\hat{n}}{c}\right)^3}$$

Suppose $\vec{v} \parallel \vec{a}$ (straight-line motion) $\Rightarrow \vec{v}\times\vec{a} = 0$

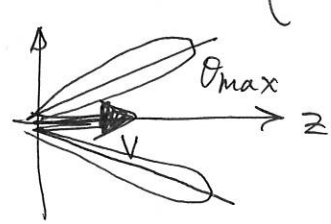
$$\hat{n}\times(\hat{n}\times\vec{a}) = \hat{n}(\hat{n}\cdot\vec{a}) - \vec{a}$$

$$\frac{dI}{d\Omega} = c\varepsilon_0\vec{E}^2R^2 = c\varepsilon_0\left(\frac{q}{4\pi\varepsilon_0c^2}\right)^2\frac{|\vec{a}-\hat{n}(\hat{n}\cdot\vec{a})|^2}{\left(1-\frac{\vec{v}\cdot\hat{n}}{c}\right)^6}$$

$$|\vec{a}-\hat{n}(\hat{n}\cdot\vec{a})|^2 = a^2 + (\hat{n}\cdot\vec{a})^2 - 2(\hat{n}\cdot\vec{a})^2 = a^2 - (\hat{n}\cdot\vec{a})^2$$

let $\vec{v} = v\hat{z} \Rightarrow \vec{v}\cdot\hat{n} = v\cos\theta$, $\vec{a}\cdot\hat{n} = a\cos\theta$

$$\frac{dI}{d\Omega} = c\varepsilon_0\frac{q^2}{16\pi^2\varepsilon_0c^3}\frac{a^2\sin^2\theta}{\left(1-\frac{v}{c}\cos\theta\right)^6}$$



For relativistic motion

$$\frac{dI}{d\Omega} = \frac{q^2}{16\pi^2\varepsilon_0 c^3}\,\frac{a^2\sin^2\theta}{\left(1-\frac{v}{c}\cos\theta\right)^6}$$

But $1-\frac{v^2}{c^2} \ll 1 \Rightarrow$ expand around $\theta = 0$:

$$\left(1-\frac{v}{c}\cos\theta\right)^6 = \left(1-\frac{v}{c}\left(1-\frac{\theta^2}{2}\right)\right)^6 = \left(1-\frac{v}{c}+\frac{v\theta^2}{2c}\right)^6 \cong \left(\frac{1}{2\gamma^2}+\frac{\theta^2}{2}\right)^6$$

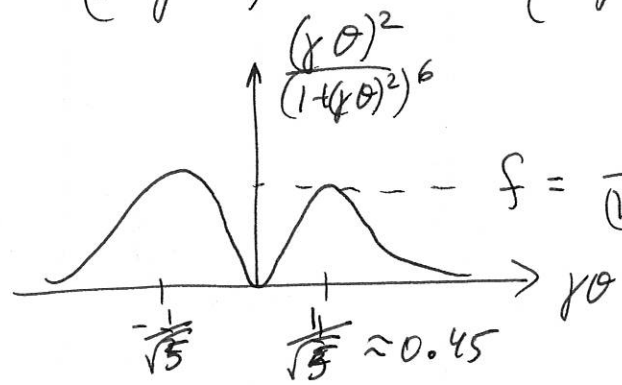
$$\gamma = \frac{1}{\sqrt{1-\frac{v^2}{c^2}}} = \frac{1}{\sqrt{1-\frac{v}{c}}\sqrt{1+\frac{v}{c}}} \approx \frac{1}{\sqrt{2}\sqrt{1-\frac{v}{c}}} \gg 1$$

$$\Rightarrow 1-\frac{v}{c} = \frac{1}{2\gamma^2}$$

$$\frac{a^2\sin^2\theta}{\left(1-\frac{v}{c}\cos\theta\right)^6} \approx \frac{a^2\theta^2}{\left(\frac{1}{2\gamma^2}+\frac{\theta^2}{2}\right)^6} = \frac{a^2\theta^2}{\left(\frac{1}{2\gamma^2}\right)^6\left(1+\gamma^2\theta^2\right)^6} =$$

$$= \frac{2^6 a^2 \gamma^{12}\theta^2}{(1+\gamma^2\theta^2)^6} = \frac{64\, a^2\gamma^{10}(\gamma^2\theta^2)}{(1+\gamma^2\theta^2)^6}$$

Notice huge enhancement factor $\gamma^{10} \gg 1$.



$$f = \frac{1/5}{(1+1/5)^6} = \frac{5^5}{6^6}$$

$$f(x) = \frac{x^2}{(1+x^2)^6} \Rightarrow f(z) = \frac{z}{(1+z)^6} \quad f'(z) = 0 = \frac{1}{(1+z)^6} - \frac{6z}{(1+z)^7}$$

$z = (\gamma\theta)^2$

$$\Rightarrow 1 = \frac{6z}{1+z} \Rightarrow 1 = 5z,\ z = 1/5$$

Max at $\gamma\theta = \frac{1}{\sqrt{5}}$, $\theta_{max} = \frac{1}{\sqrt{5}\gamma} \ll 1$

which justifies the expansion for $\gamma \gg 1$.

"Radiation friction"

Power emitted by moving particle

$$\underline{P} = \int I \, d\Omega = \frac{\mu_0 q^2}{6\pi c} \dot{v}^2 = \text{rate of energy loss}$$

$\Rightarrow$ there must be a force $\vec{F}_{rad}$ acting on q

Work by it $\int_0^{t_1} \vec{F}_{rad} \cdot \vec{v} \, dt = -\frac{\mu_0 q^2}{6\pi c} \int_0^{t_1} dt \, \dot{v}^2$

Here we make big assumption: charge is performing stationary motion (aside from P being produced) so that $-\int P \, dt$ represents <u>all</u> energy lost in time t_1

$$\int \dot{\vec{v}} \cdot \dot{\vec{v}} \, dt = \int \frac{d\vec{v}}{dt} \frac{d\vec{v}}{dt} dt = \int_0^{t_1} \frac{d\vec{v}}{dt} d\vec{v} =$$

$$= \underbrace{\left(\vec{v} \cdot \frac{d\vec{v}}{dt} \right) \Big|_0^{t_1}}_{=0} - \int_0^{t_1} \vec{v} \frac{d^2 \vec{v}}{dt^2} dt = -\int \vec{v} \cdot \frac{d\vec{a}}{dt} dt$$

$$\Rightarrow \int_0^{t_1} \left(\vec{F}_{rad} - \frac{\mu_0 q^2}{6\pi c} \dot{\vec{a}} \right) \vec{v} \, dt = 0$$

radiation reaction force (radiation friction)

$$\vec{F}_{rad} = \frac{\mu_0 q^2}{6\pi c} \frac{d\vec{a}}{dt} \quad \text{Abraham - Lorentz f-la}$$

$$= \frac{\mu_0 q^2}{6\pi c} \ddot{\vec{v}}$$

<u>Pathology</u>: $m \dot{\vec{v}} = \frac{\mu_0 q^2}{6\pi c} \ddot{\vec{v}} \Rightarrow \vec{v} = \text{const}$

free-particle

OR

$$\vec{v} = \vec{v}_0 e^{\alpha t}$$

$$m \vec{v}_0 \alpha e^{\alpha t} = \frac{\mu_0 q^2}{6\pi c} \vec{v}_0 \alpha^2 e^{\alpha t}$$

Conditions of applicability

Really, we should think of $\vec{F}_{rad}$ as a small correction.
For particle moving in external field (non-relativistically)

$$m\ddot{\vec{v}} = q\vec{E} + q\,\vec{v}\times\vec{B} + \underbrace{\frac{\mu_0 q^2}{6\pi c}\dddot{\vec{v}}}_{\text{perturbation}}$$

$$\ddot{\vec{v}} \approx \frac{q}{m}\dot{\vec{E}} + \frac{q}{m}\dot{\vec{v}}\times\vec{B} = \frac{q}{m}\dot{\vec{E}} + \left(\frac{q}{m}\right)^2 \vec{E}\times\vec{B}$$

$$\dot{\vec{v}} = \frac{q}{m}\vec{E}$$

for motion with small velocity!

$$\Rightarrow \vec{F}_{rad} = \frac{\mu_0 q^2}{6\pi c}\left(\frac{q}{m}\dot{\vec{E}} + \left(\frac{q}{m}\right)^2 \vec{E}\times\vec{B}\right)$$

For oscillatory motion $\dot{\vec{E}} \sim \omega\vec{E}$

$$\Rightarrow 1^{st}\text{ term} = \frac{\mu_0 q^3 \omega}{6\pi c m}\vec{E}$$

$$2^{nd} = \frac{\mu_0 q^4}{6\pi c m^2}\vec{E}\times\vec{B}$$

F_{rad} should be small correction to the Coulomb/Lorentz forces

$\Rightarrow F_{rad} \ll$ Coulomb force $= qE$

$$\Rightarrow \frac{\mu_0 q^2}{cm}\omega \ll 1 \Rightarrow \underbrace{\frac{c}{\omega}}_{\lambda} \gg \frac{\mu_0 q^2}{m} = \frac{q^2}{mc^2\varepsilon_0} \sim R_0$$

$\Rightarrow$ radiation friction small if wavelength $\lambda \gg R_0$

Radius of electron $\frac{e^2}{4\pi\varepsilon_0 R_0} \sim mc^2 \Rightarrow R_0 \sim \frac{e^2}{4\pi\varepsilon_0 c^2 m}$

Ex charged oscillator

$$m\dot{\vec{v}} = -k\vec{x} + \frac{\mu_0 q^2}{6\pi c}\ddot{\vec{v}}$$

$$\ddot{x} = -\frac{k}{m}x + \frac{\mu_0 q^2}{6\pi mc}\dddot{x}$$

small correction to a periodic motion

But $\dddot{x} \approx -\frac{k}{m}\dot{x} = -\omega_0^2\dot{x}$

$$\Rightarrow \quad \ddot{x} = -\omega^2 x - \frac{\mu_0 q^2}{6\pi mc}\omega_0^2\dot{x}$$

$$\ddot{x} + \underbrace{\alpha\dot{x}}_{\text{frictional force proportional to } \vec{v}!} + \omega_0^2 x = 0$$

Add external field $q\vec{E}$ in the right-hand side

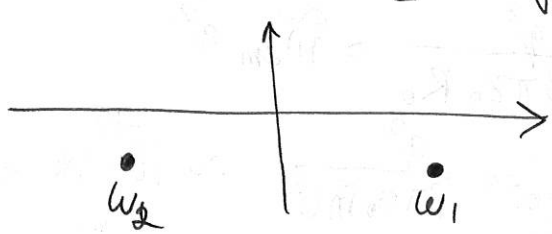
$\Rightarrow$ problem 1 from HW01, with $\gamma = \alpha \cdot m = \frac{\mu_0 q^2 \omega_0^2}{6\pi c}$

$$x(t) = -\frac{q}{m}\int \frac{d\omega}{2\pi}\frac{\vec{E}_\omega e^{-i\omega t}}{\omega^2 + i\gamma\omega - \omega_0^2}$$

Thus friction $\gamma = m\alpha$ gives us a rule for selecting contour of integration over ω.

$$\omega_{1,2} = -i\frac{\gamma}{2} \pm \sqrt{\omega_0^2 - \left(\frac{\gamma}{2}\right)^2}$$

$\mathrm{Im}(\omega_{1,2}) < 0$



"The end of classical physics"

Hydrogen atom classically:

$$E = \frac{1}{2} m v^2 - \frac{e^2}{4\pi\varepsilon_0 r} = -\frac{e^2}{8\pi\varepsilon_0 r}$$

$r_0 = 0.5\,\text{Å} \sim 10^{-10}\,\text{m}$

$$F = \frac{e^2}{4\pi\varepsilon_0 r^2} = m \underbrace{\frac{v^2}{r}}_{a} \Rightarrow m v^2 = \frac{e^2}{4\pi\varepsilon_0 r}$$

$$P = \frac{\mu_0 e^2}{6\pi c} a^2 = \frac{\mu_0 e^2}{6\pi c}\left(\frac{v^2}{r}\right)^2 = \frac{\mu_0 e^2}{6\pi c}\left(\frac{e^2}{4\pi\varepsilon_0 m r^2}\right)^2$$

But $$P = -\frac{dE}{dt} = \frac{d}{dt}\left(\frac{e^2}{8\pi\varepsilon_0 r}\right) = -\frac{e^2}{8\pi\varepsilon_0 r^2}\frac{dr}{dt}$$

$$\Rightarrow \frac{dr}{dt} = -\frac{8\pi\varepsilon_0}{e^2}\frac{\mu_0 e^2}{6\pi c}\left(\frac{e^2}{4\pi\varepsilon_0 m r}\right)^2 =$$

$$= -\frac{1}{3c}\left(\frac{e^2}{2\pi\varepsilon_0 m c}\right)^2 \frac{1}{r^2}$$

$$T = \int dt = -3c\left(\frac{2\pi\varepsilon_0 m c}{e^2}\right)^2 \int_{r_0}^{0} r^2 dr = c\left(\frac{2\pi\varepsilon_0 m c}{e^2}\right)^2 r_0^3$$

$$T \simeq 3\cdot 10^8 \left[\frac{2\pi \; 8.85\cdot 10^{-12}\cdot 9\cdot 10^{-31}\cdot 3\cdot 10^8}{(1.6\cdot 10^{-19})^2}\right]^2 (10^{-10})^3$$

$$= 3\,\pi^2\,(9^2)^2\,3^2\,10^{8-30}\,(10^3)^2$$

$$\sim 3\cdot 9\cdot 9\cdot 9^4\,10^{-16} \sim \underline{\underline{10^{-10}\,\text{s}}}$$

$\Rightarrow$ there are <u>no atoms</u>!

For a particle of mass m

$$\Delta p \cdot \Delta x \geq \frac{\hbar}{2}$$

$$\Delta x_{min} < \frac{\hbar}{\Delta p_{max}} = \frac{\hbar}{mc} = \text{"quantum radius"} \; R_q$$

Classical radius of charged particle $\frac{e^2}{4\pi\varepsilon_0 R_0} = mc^2$

$$R_0 = \frac{e^2}{4\pi\varepsilon_0 mc^2}$$

$$\frac{R_0}{R_q} = \frac{e^2}{4\pi\varepsilon_0 mc^2} \frac{mc}{\hbar} = \frac{e^2}{4\pi\varepsilon_0 \hbar c} = \frac{1}{137}$$

$$\Rightarrow R_0 \ll R_q$$

Quantum effects begin well before classical EM breaks down.